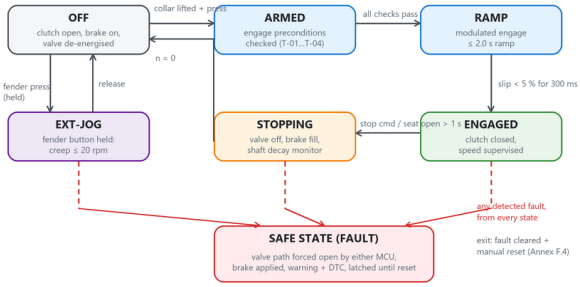
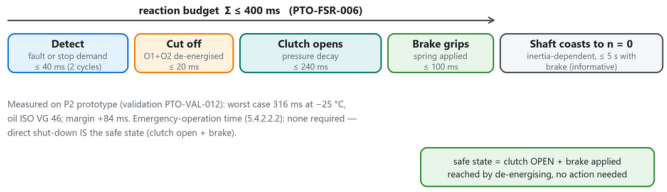
Technical_Safety_Concept

ReqID	Contents	AgPL	Allocation	Refines	Status
PTO-TSC-000	**Technical Safety Concept — Rear PTO** Technical safety requirements per ISO 25119-3:2018, Clause 5. Each TSR cross-references the functional safety requirement(s) it refines; states, times and safe-state transitions are specified per 5.4.2.2.2.	QM	System	—	Verified
—	**1. States and times (5.4.2.2.2)** **PTO control state machine (SW component SM-1)**  *Figure 1 — operating states of the PTO control; the safe state is reachable from every state* **Fault-reaction / stop chain — response-time budget (TSC §4, PTO-TSR-020)**  *Figure 2 — stop-chain response-time budget; validated worst case 316 ms*	QM	System	—	Verified
PTO-TSR-001	Cab engage command validity: NO contact closed AND NC contact open, both transitions inside a 50 ms window; any other pattern for > 100 ms raises DTC C1101 and inhibits engagement.	AgPL d	Sensor / input	PTO-FSR-001	Verified
PTO-TSR-002	Valve energisation requires L1 PWM active AND L2 enable conducting (series O1·O2). Each MCU shall be able to reach the safe state alone by opening its own switch.	AgPL d	ECU / Hardware	PTO-FSR-002, -005	Verified
PTO-TSR-003	Valve current shall be inside the command-dependent window (off: < 50 mA; hold: 380–700 mA) within 30 ms of every command change; violation triggers the stop chain (D1).	AgPL d	ECU / Hardware	PTO-FSR-004	Verified
PTO-TSR-004	Start interlock: on power-up both MCUs run BIST (RAM march, flash CRC, output pull-in test below clutch-fill pressure); outputs stay disabled until both report PASS and the engage command is absent.	AgPL d	ECU / Software	PTO-FSR-003	Verified
PTO-TSR-005	The engage state machine shall power up in OFF; the engaged state shall not be persisted (no NVM latch of engagement; parameter area write-protected in operation).	AgPL c	ECU / Software	PTO-FSR-007	Verified
PTO-TSR-006	Stop-chain timing allocation: detection ≤ 40 ms (2 application cycles), output de-energise ≤ 20 ms, clutch pressure < touch point ≤ 240 ms, brake engaged ≤ 100 ms. Budget Σ ≤ 400 ms; validated with cold oil (−25 °C).	AgPL c	System	PTO-FSR-006	Verified
PTO-TSR-007	Operator-presence processing: two seat-switch contacts sampled at 10 ms; disagreement > 500 ms = presence fault (inhibit new engagement, warning). Presence loss > 1 s in mobile mode triggers the stop chain. Stationary mode requires: PTO speed ≤ 540 mode, park brake applied, deliberate enable.	AgPL c	Sensor / input	PTO-FSR-008	Verified
PTO-TSR-008	Fender jog: accepted only in state OFF with cab switch OFF; output limited to jog pulses (creep ≤ 20 rpm); release or 10 s timeout stops within 200 ms; stuck-button (no release edge after jog end) inhibits further external engagement (DTC C1108).	AgPL b	System	PTO-FSR-009, -010	Verified
PTO-TSR-009	Speed-mode supervision: coded lever position vs measured n_ENG/n_PTO ratio; mismatch > 500 ms inhibits engagement (standstill) or triggers DM-2 (running).	AgPL b	Sensor / input	PTO-FSR-011, -012	Verified
PTO-TSR-010	Shaft-speed supervision: n_PTO > 1.10 × mode nominal for > 300 ms triggers DM-2 controlled disengage (ramp-down ≤ 1 s) and warning.	AgPL b	ECU / Software	PTO-FSR-011	Verified
PTO-TSR-011	Engagement ramp: closed-loop current ramp 0 → hold in ≤ 2.0 s; slip (n_PTO vs expected) supervised; ramp abort on stall or overcurrent.	AgPL b	ECU / Software	PTO-FSR-013	Verified
PTO-TSR-012	Cross-monitoring: SPI question-answer every 20 ms with 16 pre-computed challenges; two consecutive misses = channel fault → safe state (D4). External window watchdog (U6) supervises L1 with 25–75 ms window; L2 has an independent internal WD.	AgPL d	ECU / Hardware	PTO-FSR-002, -004	Verified
PTO-TSR-013	Supply supervision: 5 V / 3.3 V window monitors reset both MCUs outside 4.5–5.5 V / 3.0–3.6 V; the output stage is disabled during reset and brown-out (outputs pulled to safe by hardware).	AgPL d	ECU / Hardware	PTO-FSR-004	Verified
PTO-TSR-014	Environmental withstand: ISO 15003 cab class (−25…+70 °C, 5 g vibration), ISO 7637-2 pulses 1/2a/3/5a (clamped load dump), ISO 14982 EMC, reverse polarity −14 V, IP54 connector zone.	AgPL d	ECU / Hardware	all	Verified
PTO-TSR-015	Manual reset per Annex F.4: after a latched safe state, recovery requires fault absence AND cab switch to OFF AND a deliberate reset action (switch held OFF 2 s); reset shall not start motion by itself.	AgPL c	ECU / Software	PTO-FSR-004	Verified
PTO-TSR-016	An overspeed reaction shall use the controlled disengage ramp (not an instantaneous cut) unless a second, higher-priority stop demand is present; abrupt full-load cuts are themselves a driveline hazard (PTO-HZ-08).	AgPL b	ECU / Software	PTO-FSR-011, -013	Verified
PTO-TSR-017	Indication: PTO state and warnings on CAN with alive counter + checksum; loss of indication shall not affect any safety function (QM path, PTO-FSR-014).	QM	System	PTO-FSR-014	Verified
PTO-TSR-018	Safety-related parameters (mode nominal speeds, ramp table, current windows) are stored twice with CRC-16 each; a mismatch or range violation at start-up inhibits engagement (parameterisation rules per Part 3, 7.8).	AgPL c	ECU / Software	PTO-FSR-003, -011	Verified